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Dual-functional additives enabling a high-performance silicon–carbon composite anode in pouch cells at high temperatures†

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Silicon–carbon composite (Si–C) is currently the most promising anode material for lithium-ion batteries (LIBs) due to its high energy density. However, the large volume change of Si particles during cycling causes sustained damage to the solid electrolyte interface (SEI), leading to deterioration of cycling performance, especially at high temperatures. In this study, fluoroethylene carbonate (FEC) and succinic anhydride (SA) were employed as dual-functional additives to improve the cycling performance of the Si–C anode at high temperatures. High-energy-density $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ (NCM811)/(Si–C) pouch cells were assembled to evaluate the performance of the dual-functional additives. It is found that an appropriate amount of FEC improved the cycling stability of pouch cells at room temperature by facilitating the formation of LiF-rich SEI. However, this improvement failed at high temperatures due to the triggering of massive side reactions. The incorporation of both FEC together and SA effectively solved this problem and improved the cycling and storage performance of pouch cells at high temperatures by forming a uniform and dense Li_2CO_3 -rich SEI film, which provided good thermal stability. Density functional theory calculations and ultrasound transmission imaging were used to illustrate the mechanism. All data were collected at pouch cell level, making the findings more persuasive.

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1 Introduction

With the rapid growth of electric vehicles and energy storage systems, lithium-ion batteries (LIBs) have become increasingly important in daily life. To meet growing market requirements, LIBs are designed not only to provide higher energy density but also to adapt to complex conditions, such as high-temperature environments, ensuring good cycle stability and high safety.^{1,2} Traditional graphite (Gr) anodes cannot fully satisfy future requirements because of their limited theoretical capacity (372 mA h g^{-1}). Silicon (Si) has emerged as the most promising anode material for high-energy LIBs because of its high theoretical capacity (4200 mA h g^{-1}), low working voltage, abundant resources, and environmental friendliness.^{3–6} Therefore, combining a high-capacity Si-based anode with a layered oxide $\text{LiNi}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}\text{O}_2$ (NCM811) cathode is an efficient strategy for achieving high energy density in LIBs. However, the large-

scale commercial application of Si-based anodes is hindered by problems such as poor conductivity, low coulombic efficiency, rapid capacity decay, pulverization of the electrode, and unstable SEI film caused by substantial volume change during the charging/discharging process.^{7–12}

To solve the above problems, several practical strategies have been proposed, including designing nanostructured silicon,^{13–15} developing silicon–carbon composites,^{11,16–18} alloying silicon with metals,^{19–21} regulating the interface structure,^{10,22,23} forming surface films,^{24,25} conducting unconventional electrode engineering,^{26,27} and developing advanced binders.^{28–31} Besides, electrolyte optimization is also a promising way to enhance the performance of Si-based anodes as it determines the charge–discharge coulombic efficiency, internal impedance, ion transport, cycle life, and safety performance of the batteries.^{32–36} The above methods carry unique benefits and constraints and enhance LIB performance in practical applications.

In recent years, the incorporation of additives into bulk electrolytes has emerged as one of the most cost-effective and scalable ways to improve the performance of Si-based anodes. Fluoroethylene carbonate (FEC),^{37–43} vinylene carbonate (VC),^{44–49} pentafluorophenyl isocyanate (PFPI),⁵⁰ propargyl-diethylthiophosphate (PDTP),⁵¹ and lithium oxalyl-di-fluoroborate (LiDFOB)⁵² have been reported as potential additives for Si-based anodes. The electrolyte additives enable the Si-based anodes to create a stable SEI to accommodate the volume

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expansion of Si.⁵³ For example, Yang *et al.*⁵⁴ designed an electrolyte based on glycol dimethyl ether groups (GlyEs) to ensure conformal SEI on Si particles. This SEI structure with a robust outer layer and an elastic inner layer can reduce the internal resistance and polarization, thereby improving the cycling performance of the battery. Li *et al.*⁵⁵ compared the SEI composition on Si–C composite anodes with and without an FEC additive and found that a denser SEI was formed in an FEC-containing electrolyte, blocking small molecules and inhibiting LiPF₆ hydrolysis. The LiF produced by the decomposition of FEC effectively covers Si particles and hence enhances cycling stability.

Individual electrolyte additives have their advantages and disadvantages. For example, FEC also introduces the problem of gas production and rapid consumption at high temperatures. The synergy of multiple additives has been recognized as a promising strategy for further improving the performance of Si–C anodes.^{52,56} Aupperle *et al.*⁵³ reported (2-cyanoethyl)triethoxysilane (TEOSCN) as a synergistic additive for FEC/VC containing electrolyte and found that a robust electrode/electrolyte interface was formed by $\text{C}\equiv\text{N}$ reduction/oxidation on the surface of NMC cathode and Si–Gr anode particles, which prevented the dissolution of NMC active species and avoided the formation of by-products, thereby improving the high-temperature performance.

In this study, we investigated the effect of FEC concentration on the cycling performance of NCM811/Si–C pouch cells, especially at high temperatures. Adding an appropriate concentration of FEC can enhance the room-temperature cycling performance of the cells by forming a stable SEI-rich LiF, but there is a notable decline in the high-temperature performance due to the poor thermal stability of FEC. Consequently, succinic anhydride (SA) was introduced simultaneously as a dual-functional additive to improve the high-temperature performance. With only 0.8% SA added to the FEC-containing electrolyte, cycling performance at 45 °C was significantly improved, which is attributed to the formation of Li₂CO₃/R–O–CO₂–Li rich, uniform and dense SEI induced by SA at high temperatures. A series of experiments and simulations, including scanning electron microscopy (SEM), transmission electron microscopy (TEM), X-ray photoelectron spectroscopy (XPS), ultrasound transmission imaging (UTI) and density functional theory (DFT), were conducted to discover the mechanism of the enhancement.

2 Experimental section

2.1 Preparation of electrolyte solutions

Battery-grade organic solvents including ethylene carbonate (EC, ≥99.5%), ethyl methyl carbonate (EMC, ≥99.5%), diethyl carbonate (DEC, ≥99.5%), lithium hexafluorophosphate (LiPF₆, ≥99.7%), fluoroethylene carbonate (FEC, ≥95%), and succinic anhydride (SA, ≥95%) were provided by Canrd New Energy Technology Co., Ltd, China. 1 mol per L LiPF₆ was dissolved in a mixture of EC, EMC and DEC at a volume ratio of 1 : 1 : 1 to prepare a blank electrolyte, and FEC was dissolved in the blank electrolyte with a mass ratio of 5%, 8% and 12%. SA with mass

ratios of 0.4%, 0.8% and 1.2% was added to an electrolyte containing 8% FEC to obtain dual-functional additives containing electrolytes.

2.2 Preparation of pouch cells

Sealed and dry machine lamination pouch cells were used for long-term electrochemical evaluation of the electrolyte formulations. For the LiNi_{0.8}Mn_{0.1}Co_{0.1}O₂ (NCM811)/Si–C pouch cell, the cathode was prepared by mixing 97% NCM811, 1.0% polyvinylidene fluoride (PVDF), 1.1% carbon nanotubes (CNT), and 0.9% conductive carbon black (in mass ratio) into a homogeneous slurry with *N*-methyl pyrrolidone, followed by coating on aluminum foil with a thickness of 10 μm. The slurry was coated on a single side and dried under vacuum at 80 °C for 12 h. The cathode plate was cut into pieces with a width of 53.5 mm and length of 76 mm, and the cathode pieces were welded with an aluminum lug for later use.

The anode was prepared by mixing 50% artificial graphite, 40% SiO_x, 1.5% carboxymethyl cellulose (CMC), 1% polyacrylic acid (PAA), 2.2% styrene–butadiene rubber (SBR), and 5.3% conductive carbon black (in mass ratio) into a homogeneous slurry with deionized water, followed by coating on copper foil (thickness of 6 μm). The slurry was coated on a single side and dried in a vacuum at 80 °C for 12 h. The anode plate was cut into pieces with a width of 56.5 mm and a length of 79 mm, and the anode pieces were welded with a nickel lug for later use. The winding method was used to prepare pouch cells using polypropylene (PP) as the separator. All chemical reagents were provided by Canrd New Energy Technology Co., Ltd, China. The NCM811/Si–C pouch cells were assembled with a cathode/anode capacity ratio of 1.05 (N/P = 1.05), and the theoretical capacity of the whole cell was 3.5 Ah at the charging cut-off voltage of 4.25 V.

2.3 Electrochemical measurements

Galvanostatic charge–discharge experiments were conducted on a battery testing system (Neware Tester, Shenzhen, China) with a voltage range of 2.8–4.25 V. The cells were tested for long-term cycling performance at 25 °C and 45 °C at 1C.

Electrochemical impedance spectroscopy (EIS) measurements were performed on a CHI604E instrument (Chenhua, China). The battery was tested over a frequency range of 10 000–0.01 Hz at a voltage amplitude of 5 mV.

The high-temperature storage performance of NCM811/Si–C pouch cells was evaluated by storing them at 60 °C for 77 days. The battery's remaining capacities were tested for 7, 35, 49, 63, and 77 days. Before storage, the pouch cell was cycled three times at room temperature, and the initial discharge capacity C_0 was recorded. The fully charged battery was stored in an incubator at 60 °C for the specified period. After storage, the battery was left at room temperature for 2 h and then cycled again, with the first discharge capacity C_1 and the second discharge capacity C_2 recorded. Therefore, the residual capacity ratio $C_{R_1} = C_1/C_0$, and the capacity recovery ratio $C_{R_2} = C_2/C_0$.

2.4 Materials characterization

To observe the surface morphology and microstructure of the electrodes after cycling, the batteries were carefully disassembled in an argon-filled glove box. The anodes and cathodes were washed with DMC three times to remove residual electrolytes and then dried in a glove box at room temperature for 24 h. The phases of SiO_x and NCM811 were characterized by X-ray diffraction (XRD, D/MAX-C, Rigaku) at a 2θ range of $10\text{--}80^\circ$ with $\text{Cu K}\alpha$ radiation. The chemical components of SEI formed on the anodes were investigated using X-ray photoelectron spectroscopy (XPS, Thermo Scientific Escalab 250Xi). The morphologies and microstructures of the cathode and anode materials were examined using scanning electron microscopy (SEM, JSM7001F) and transmission electron microscopy (TEM, JEOL JEM-2100 F). The ultrasonic transmission imaging of the pouch cell was conducted using a TOPS-RD60C (Wuxi Topsound Technology Co. Ltd, China).

2.5 Computations

The molecular structures of EC, EMC, FEC, and SA were optimized *via* density functional theory (DFT) using Gaussian 09 software. Both the structural optimization and energy computations utilized the B3LYP functional and def2-TZVP basis sets.⁵⁷ The frontier molecular orbital energies of each molecule were calculated at the def2-TZVP level. Molecular orbitals were visualized using Multiwfn and VMD software.⁵⁸

3 Results and discussion

XRD patterns of SiO_x and NCM811, along with digital photographs of the corresponding pouch cells used in this study, are shown in Fig. S1.† The detailed specifications of the pouch cells

are presented in Table S1.† The SEM images and the corresponding EDS mappings of the Si-C anode and NCM811 cathode before cycling are shown in Fig. S2 and S3.† To explore the effect of FEC concentration on the cycling performance of Si-C anode at room temperature, galvanostatic charge-discharge experiments were carried out on NCM811/Si-C pouch cells. In the voltage range of 2.8–4.25 V, the pouch cell was charged and discharged at a current density of 1C until the capacity of the battery decayed to 80%. As shown in Fig. 1a, there is a notable difference in capacity retention with increasing FEC content. With capacity retention to 80%, the pouch cells with 5%, 8%, and 12% FEC run for 359, 546, and 513 cycles, respectively. The initial charge-discharge curves look similar (Fig. 1b–d). However, after 350 cycles, the pouch cell with 5% FEC experienced the largest decrease in the discharge voltage plateau, whereas the pouch cells with 8% and 12% FEC showed no obvious change. The capacity retentions after 350 cycles are 82.0%, 89.7%, and 89.1% for the pouch cells with 5%, 8%, and 12% FEC, respectively, indicating that the addition of FEC can improve the cycle performance at room temperature, which content is also critical for the improvement.

To further investigate the capacity degradation mechanism at room temperature, EIS tests were performed on the pouch cells after 450 cycles at 1C. As shown in Fig. 1e, the Nyquist plots consist of two semicircles located in the high- and middle-frequency ranges. These semicircles are associated with the resistance of SEI (R_{SEI}) and charge transfer resistance (R_{ct}), respectively. The sloped line in the low-frequency region represents the diffusion of lithium ions (W_s). The equivalent circuit of the Nyquist plots is shown in Fig. S4,† and the fitted R_{SEI} and R_{ct} values are shown in Fig. 1f. The R_{SEI} and R_{ct} values of the pouch cell with 5% FEC are notably higher than those of the other two cells. These observations suggest that low FEC

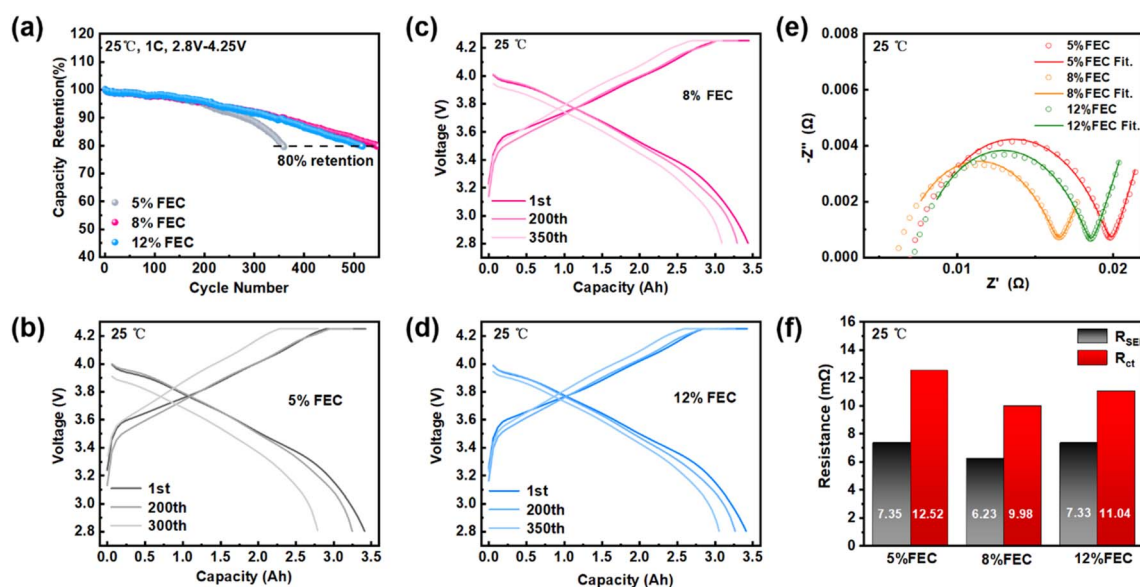


Fig. 1 (a) Cyclic performance of the NCM811/Si-C batteries with different electrolytes. Charge-discharge curves of the NCM811/Si-C batteries with (b) 5%, (c) 8% and (d) 12% FEC electrolyte at 25 °C. (e) EIS fitting results of the NCM811/Si-C battery with electrolytes of different FEC concentrations at 25 °C. (f) Fitted values of the resistance of the SEI (R_{SEI}) and charge transfer resistance (R_{ct}) at 25 °C.

contents in electrolytes are not sufficient to enhance the SEI. Consequently, an uneven SEI film diminishes the charge transfer at the interface, adversely affecting the cycling life of the cell. Conversely, a high FEC content (12%) leads to an increase in R_{SEI} and R_{ct} . This finding indicates that a higher FEC content may prompt the formation of a thicker SEI layer due to excessive side reactions, which impede the diffusion of ions. Therefore, only an optimal FEC content of around 8% can achieve a balance between inhibiting side reactions and facilitating the formation of a stable SEI film, which effectively promotes the diffusion of lithium ions and enhances the cycling performance of the battery.

Given that the SEI is greatly influenced by the FEC content, an XPS analysis was performed to investigate the chemical composition of the SEI on the Si-C anode. Fig. 2 shows the high-resolution C 1s, F 1s, and P 2p spectra of the cycled Si-C composite anodes with different FEC contents. The C 1s spectra comprise four peaks: C-C (284.8 eV), C-O (286.9 eV), R-O-CO₂-Li (288.7 eV), and Li₂CO₃ (290.0 eV).^{40,59} The pouch cells with different FEC contents showed nearly the same C 1s spectra, indicating that the contents of the C-containing species of the three SEIs were almost identical. However, by comparing F 1s and P 2p spectra, it can be found that the peak intensities of LiF

(684.5 eV) in the F 1s spectra and Li_xPF_yO_z (133.0 eV) in the P 2p spectra increased significantly with increasing FEC content.⁶⁰ The formation of Li_xPF_yO_z resulted from the decomposition of lithium salt in carbonate-based electrolytes and the electrochemical reduction process.⁵² This finding indicates that the addition of FEC can help to generate a large number of F-containing compounds and hence form LiF-rich SEI films, which are mechanically stable due to the high elastic modulus of LiF (≈ 65 GPa).⁶¹ However, a higher FEC content will result in a denser SEI with higher LiF content, lowering the R_{ct} while increasing the interfacial impedance due to the electron-insulating nature of LiF, which agrees well with the EIS results. Thus, by adjusting the LiF content to an appropriate level of around 8%, a balance between high mechanical stability and fast charge transfer of the SEI can be achieved, thereby improving the room-temperature cycle performance of the Si-C anode.

Since the cycling performance of the pouch cell with 5% FEC at room temperature was unsatisfactory, we only tested the pouch cells with 8% and 12% FEC at 45 °C. As shown in Fig. S5,[†] as the FEC content increased, the capacity retention decreased significantly at 45 °C because of the insufficient thermal stability of the FEC. Succinic anhydride (SA) exhibits better

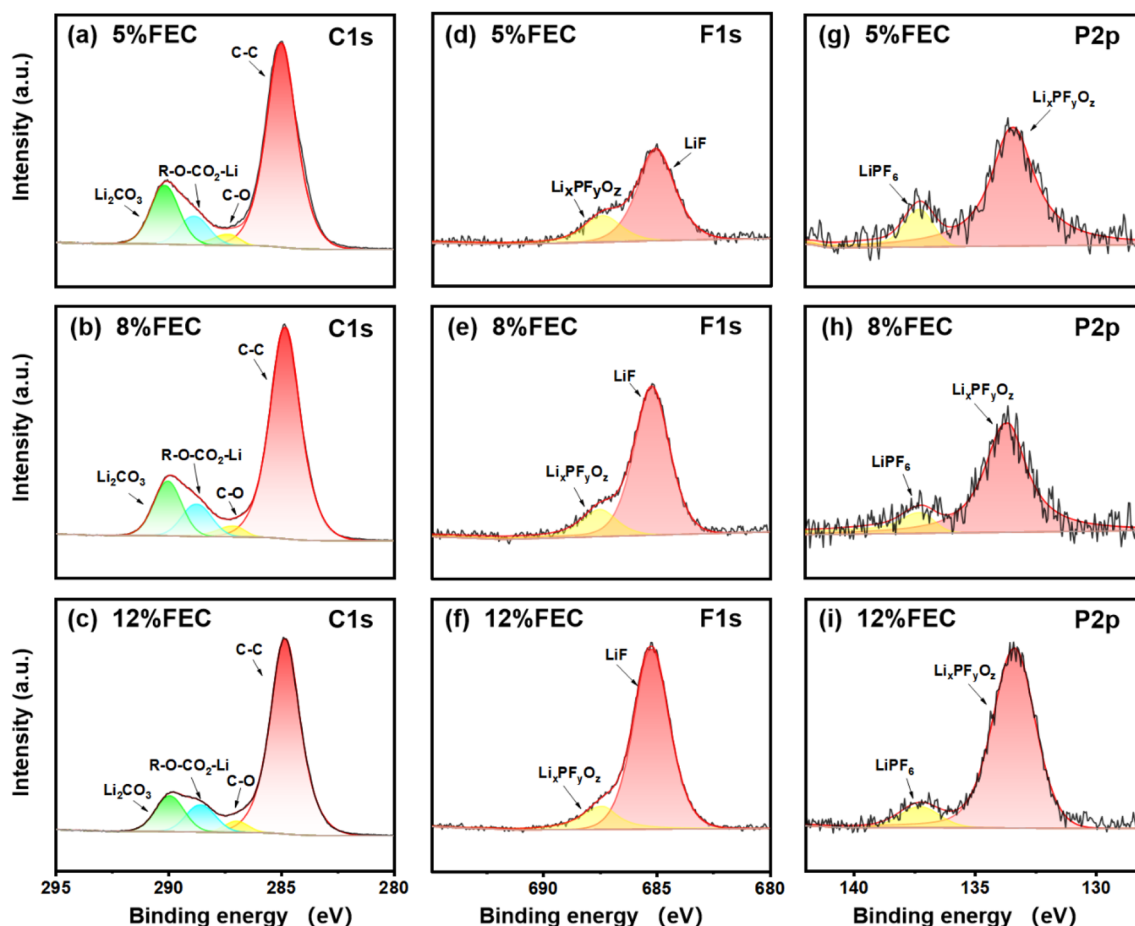


Fig. 2 C 1s, F 1s, and P 2p XPS spectra of the Si-C composite anodes with (a–c) 5% FEC, (d–f) 8% FEC and (g–i) 12% FEC electrolytes after 350 cycles.

thermal stability than FEC.⁶² Consequently, SA was introduced as a dual-additive together with FEC to further improve the high-temperature cycling stability of NCM811/Si-C pouch cells.

The electrolyte with 8% FEC was chosen as the blank control sample due to its best performance at room temperature. On this basis, SA was added to the electrolyte at mass fractions of 0.4%, 0.8% and 1.2%. The pouch cell using 8% FEC + 0.8% SA exhibited the best cycle performance at 45 °C (Fig. 3a). The 8% FEC cells with an additional SA content of 0, 0.4%, 0.8% and 1.2% showed cycle lives of 473, 529, 550 and 433, respectively, when the capacity retained 80%. With 0.8% SA as the dual additive, the 8% FEC pouch cell realized a nearly 16% increase in the cycle life. Furthermore, the combination of 8% FEC and 0.8% SA demonstrates superior performance compared with other electrolyte additives for Si-based anodes (Table S2†). However, the addition of SA up to 1.2% is not beneficial for improving cyclability, probably due to severe side reactions. To further investigate the influence of SA, we compared the charge–discharge curves of the pouch cell containing “8% FEC + 0.8% SA” (cell A) with the blank pouch cell containing only 8% FEC (cell B) (Fig. 3b and e). Their curves for the first cycle were similar. However, after the 200th and 450th cycles, the voltage plateau in cell A was higher than that in cell B. After 450 cycles, the capacity retention of cell A was 83.33% and that of cell B was

81.44%. This indicates that the introduction of SA in the electrolyte helps reduce the polarization of the pouch cell and improves cycling performance.

To gain a deeper understanding of the capacity fade mechanism during cycling, the evolution of the differential capacity (dQ/dV) of NCM811/Si-C pouch cells with different additives was investigated. As shown in Fig. 3c and f, despite the electrolyte difference, both cells typically undergo the same electrochemical processes but vary in the decay rate of their characteristic intercalation/alloying peaks' intensity. In the first cycle, both cells exhibited the same oxidation peaks at approximately 3.6, 3.7, 3.96 and 4.17 V, as well as the same reduction peaks at approximately 3.4, 3.5, 3.8 and 4.0 V. For cell B, the oxidation peak in the low-voltage region shifts from 3.57 to 3.67 V and experiences a dramatic intensity decrease after 450 cycles. This was attributed to the loss of Li at the anode side caused by the pulverization of Si particles and the continuous formation of the damaged SEI.⁶³ However, for cell A, the same peak shifted from 3.57 to 3.64 V, accompanied by a small intensity decrease. This indicates that the introduction of 0.8% SA effectively protected the Si particles and resulted in a more stable SEI, consequently alleviating the loss of Li. In the high-voltage region, the oxidation peaks at 3.96 V and 4.17 V in cell B undergo an obvious left shift and a decrease in intensity upon

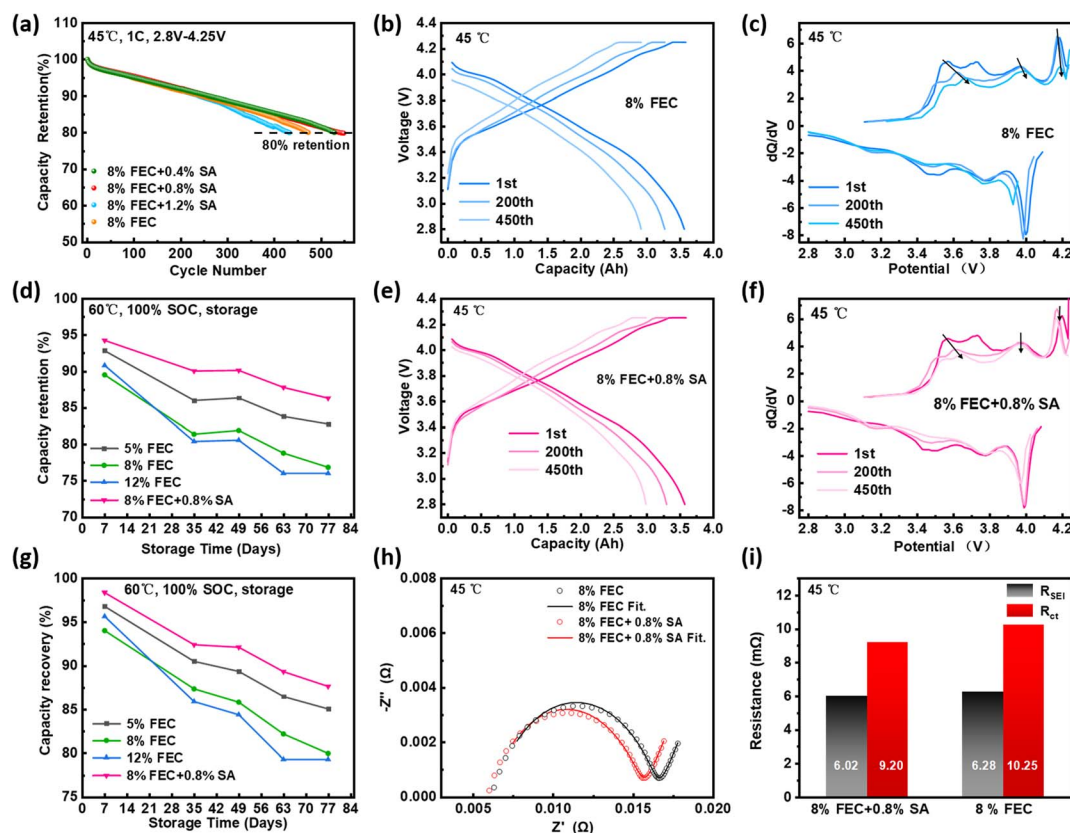


Fig. 3 Electrochemical performance of NCM811/Si-C battery at high-temperature. (a) Cycling performance of pouch cells with different electrolytes at 45 °C. (b and e) Charge–discharge curves and (c and f) dQ/dV plots of NCM811/Si-C pouch cells with 8% FEC and 8% FEC + 0.8% SA at 45 °C. (d) Capacity retention and (g) capacity recovery changes of 100% SOC batteries stored at 60 °C for 77 days. (h) EIS fitting results of the NCM811/Si-C battery with different electrolytes at 45 °C. (i) Fitted R_{SEI} and R_{ct} resistances at 45 °C.

cycling. These changes are caused by the M-H2 and H2-H3 phase transitions occurring in the cathode, respectively.⁶⁴ On the contrary, the same peaks of cell A show almost no changes during cycling, indicating that the structural degradation and resultant capacity fade of NCM811 are also alleviated by the addition of 0.8% SA at high temperatures.

A high-temperature battery storage experiment was conducted to investigate the failure modes of pouch cells with different electrolyte additives. Fig. 3d and g show the capacity retention and recovery rate of the pouch cells stored at 60 °C for 77 days. The decrease in the capacity retention and recovery rate of the cells is caused by self-discharge and irreversible reactions that occur inside the battery during high-temperature storage. For cell A, the capacity retention was 86.32%, and the recovery rate was 87.64% after 77 days, which is higher than those of the other cells. This indicates that the addition of 0.8% SA is beneficial for forming a thermally stable SEI and reducing side reactions, thereby alleviating self-discharge. Furthermore, pouch cells with higher FEC contents exhibited a much larger decrease in the capacity retention and recovery rate compared

with cells with lower FEC contents. This confirms that the decomposition of thermally unstable FEC results in the consumption of more active lithium ions and triggers severe side reactions, leading to serious irreversible capacity loss.

To investigate the mechanism by which SA enhances the thermal stability of the Si-C anode, EIS tests were performed on pouch cells after 450 cycles at 45 °C. In Fig. 3h, cell A and cell B show similar profiles. The equivalent circuit of the Nyquist plots is provided in Fig. S4,[†] and the fitted R_{SEI} and R_{ct} values are shown in Fig. 3i. The R_{SEI} and R_{ct} of cell A were both lower than those of cell B, indicating that introducing SA can reduce the side reactions at the interface by forming a thermally stable SEI layer.

To further investigate the effect of FEC-SA dual additives on the structure and morphology of the SEI on Si-C composite anode after high-temperature cycling, the pouch cells were disassembled after 450 cycles at 45 °C and were characterized using SEM and TEM. The optical photographs of the disassembled pouch cells are shown in Fig. S6.[†] In Fig. 4, the Si-C composite anode surface of cell A is smoother and more

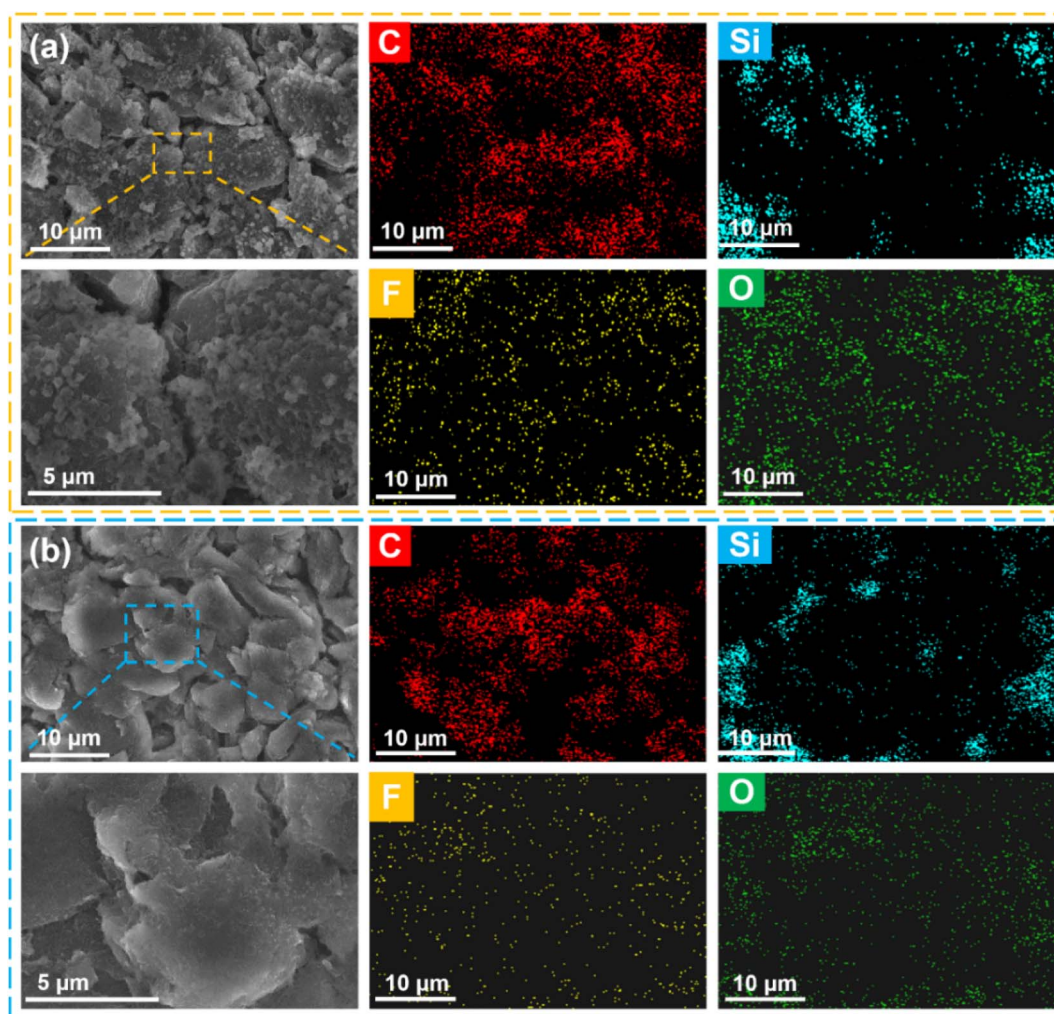


Fig. 4 SEM images and corresponding EDS maps of Si-C anodes in (a) 8% FEC and (b) 8% FEC + 0.8% SA electrolytes after 450 cycles at high temperature.

uniform than that of cell B. Furthermore, the anode EDS maps of cell A demonstrate weaker signals of F and O than those of cell B, indicating suppressed electrolyte decomposition in cell A. These findings revealed that side reactions are more violent at high temperatures when FEC is used alone. In contrast, using FEC and SA as dual additives, a uniform and dense SEI film can be formed on the surface of the Si-C anode, which effectively withstands the volume change and prevents further reactions between the Si-C anode and the electrolyte, consequently improving the high-temperature cycling performance. Furthermore, the cathode of the pouch cell using the dual additives also shows a smoother and more uniform surface, demonstrating that the dual additives may also help to form a stable cathode electrolyte interface (CEI) at high temperatures (Fig. S7†).

We used TEM to further investigate the detailed structure of the SEI on the Si-C composite anode after high-temperature cycling. As shown in Fig. 5, consistent with the SEM results, the SEI of Si-C anode for cell B appears uneven, fluffy and non-dense, which is caused by the decomposition of FEC and violent side reactions during high-temperature cycling. In contrast, the cells containing the dual additives presented a dense and uniform SEI with a thickness of 13 nm. This thin but stable SEI is beneficial for improving the charge transfer at the interface and protecting Si particles from cracking, leading to better cycling performance, which is consistent with previous electrochemical results. These findings validate that the incorporation of SA-FEC dual additives effectively promotes the formation of a uniform and dense SEI during high-temperature cycling and consequently leads to significant performance

improvement at high temperatures for NCM811/Si-C pouch cells.

We further used XPS to examine the chemical composition of the SEI formed at high temperatures. Fig. 6 shows the XPS spectra of C 1s, F 1s, and Li 1s for the Si-C anodes with different electrolytes. The Li 1s spectra were resolved into three peaks: LiF (56.0 eV), Li_2CO_3 (55.5 eV), and R-O-CO₂-Li (55.0 eV).⁶⁵ The P 2p spectra are shown in Fig. S8.† The intensities of Li_2CO_3 (289.9 eV) and C-O (286.5 eV) in C 1s for cell A are much higher than those for cell B, demonstrating that the Li_2CO_3 content in the SEI increases after SA addition. Furthermore, the intensity of LiF (685.0 eV) in F 1s decreased while the intensities of Li_2CO_3 (55.5 eV) and R-O-CO₂-Li (55.0 eV) in Li 1s increased, suggesting that the LiF content in the SEI formed by the dual additives decreased while the Li_2CO_3 and R-O-CO₂-Li contents increased. Previous studies have reported that a high content of alkali carbonates enhances the thermal and cycling stability of SEI.^{62,66,67} Thus, the FEC-SA dual additives can facilitate the formation of a uniform and dense Li_2CO_3 /R-O-CO₂-Li rich SEI and hence effectively address the limitation of the FEC single additive in improving high-temperature cycling performance. Additionally, the trace amount of SA in the electrolyte did not adversely affect the cycling performance of the pouch cells at room temperature (Fig. S9†).

Ultrasound transmission imaging with high sensitivity to gas/vacuum interfaces was performed to monitor gas production and electrolyte consumption in pouch cells containing different electrolytes after 35 days of storage at 60 °C. The peak-to-peak amplitude values (PPV) of the recorded signals were

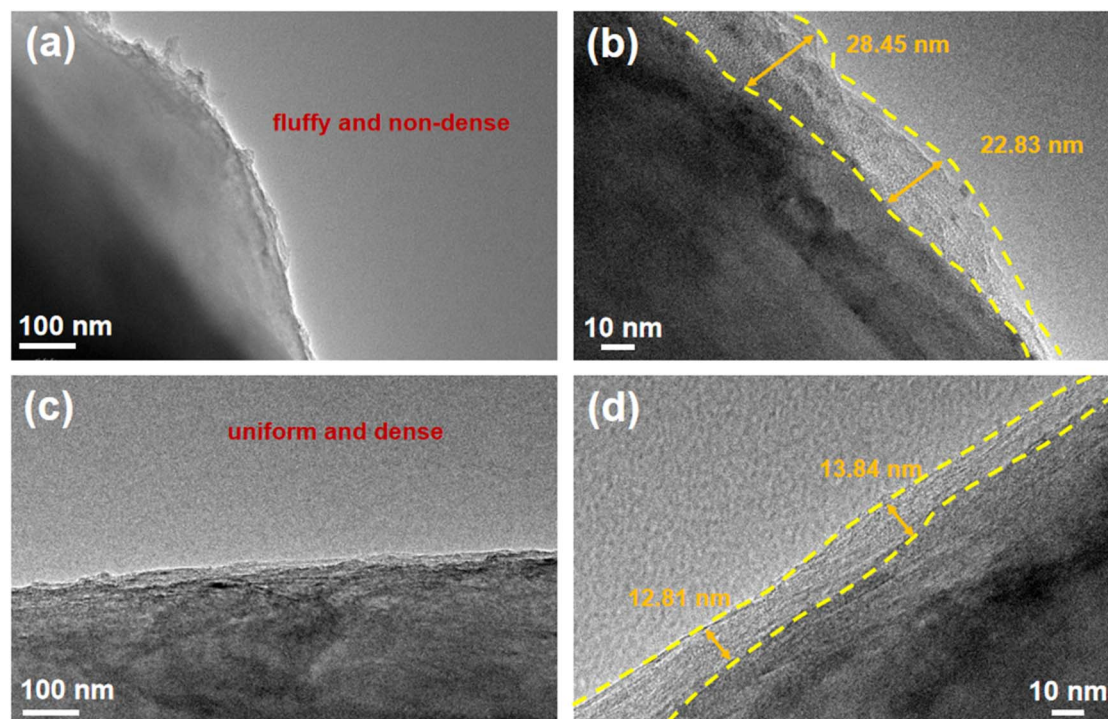


Fig. 5 TEM images of the Si particles/Si-C anodes of the pouch cells with (a and b) 8% FEC and (c and d) 8% FEC + 0.8% SA electrolyte after 450 cycles at high temperature.

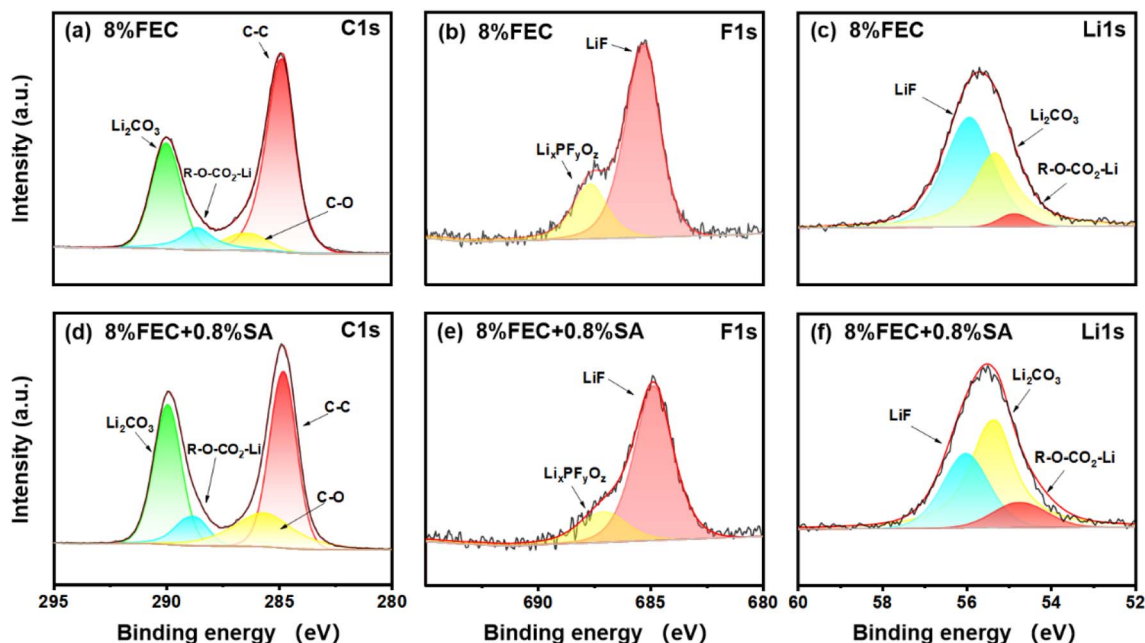


Fig. 6 C 1s, F 1s, and Li 1s XPS spectra of the Si-C composite anodes with (a–c) 8% FEC and (d–f) 8% + 0.8% SA electrolytes after 450 cycles.

converted into a blue-to-red color gradient to generate ultrasonic images. Gas formation and electrolyte depletion caused by side reactions during repeated SEI formation lead to bluer regions in the images as signal attenuation intensifies at solid–air or liquid–air interfaces.^{68,69} As shown in Fig. 7, the pouch cell with 12% FEC exhibited the largest blue area, followed by the cells with 8% FEC and 5% FEC. In contrast, the pouch cell with FEC-SA dual additives exhibited the smallest blue area, suggesting that the dual-additive system promoted a stable SEI layer and significantly suppressed parasitic reactions.

DFT calculations were performed to better illustrate the mechanism of FEC-SA dual additives in modulating the SEI structure. The reduction decomposition potential of organic

molecules is related to their lowest LUMO energy. As shown in Fig. 8a and b, FEC and SA exhibit the lowest LUMO energy compared with the solvents (DEC, EMC and EC), indicating that the additives are more reducible than the solvents. The working mechanism of the SA and FEC is shown in Fig. 7c. The LUMO energy of SA (−1.176 eV) is much lower than that of FEC (0.162 eV), indicating that even a trace of SA can be reduced more effectively than FEC. Consequently, the preferential decomposition of SA leads to the formation of $\text{Li}_2\text{CO}_3/\text{R-O-CO}_2\text{-Li}$ -rich SEI at high temperatures, as supported by TEM and XPS. The Li_2CO_3 -rich SEI delivered outstanding thermal stability, thereby improving the high-temperature cycling performance of the pouch cells.

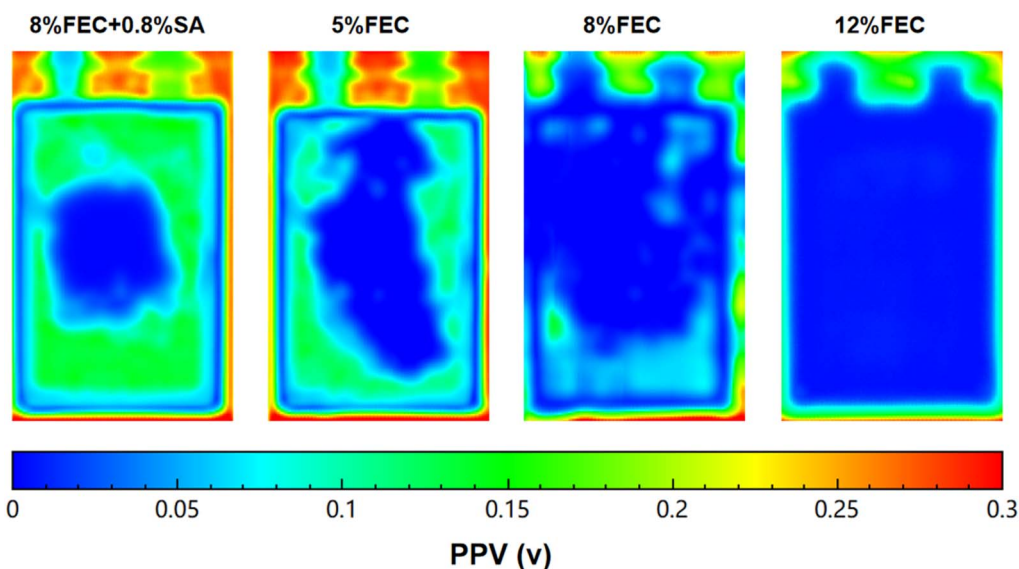


Fig. 7 Ultrasonic transmission imaging of pouch cells with different electrolytes after storage at 60 °C for 35 days.

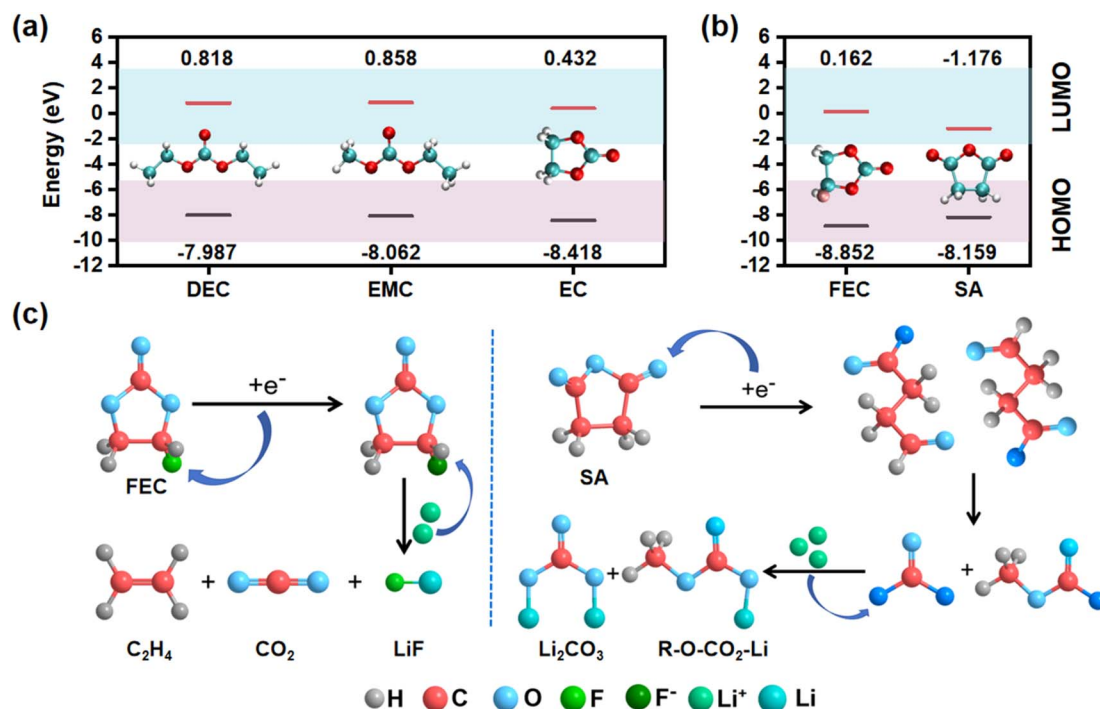


Fig. 8 Calculated $E_{\text{HOMO}}/E_{\text{LUMO}}$ energies of (a) DEC, EMC, EC and (b) FEC, SA. (c) Schematic illustration of the working mechanism of FEC and SA in modulating the SEI structure.

4 Conclusions

In summary, the effect of the dual FEC-SA additives on the cycling performance at high temperatures and interfacial chemistry of Si-C composite anode was systematically investigated at the level of the pouch cells. The addition of FEC to the electrolyte results in the formation of a LiF-rich SEI film on the Si-C anode surface. As demonstrated by our experiments, the optimal 8% FEC content in the electrolyte can effectively improve the cycling stability of NCM811/Si-C pouch cells at room temperature, but it fails at high temperatures. However, the introduction of small SAs significantly improved the high-temperature cycling performance. The pouch cell using “8% FEC + 0.8% SA” dual additives achieve 80% capacity retention after 550 cycles at 45 °C and maintained a high-capacity retention and recovery rate above 86% after storage at 60 °C for 77 days, which is attributed to the formation of Li_2CO_3 -rich SEI with high thermal stability. This study further demonstrated the effectiveness of the dual FEC-SA additives in regulating the SEI at high temperatures.

Data availability

The data supporting this article have been included as part of the ESI.†

Author contributions

Xiaohe Hou: data curation, formal analysis, investigation, methodology, visualization, writing – original draft. Jianlong

Cong: software, investigation. Shiwen Gong: investigation. Zezhuo Li: investigation. Yuxuan Shao: investigation. Aiqi He: investigation. Yuxin Fan: investigation, conceptualization, formal analysis, methodology, visualization, supervision, project administration, writing – review & editing. Yunhui Huang: supervision, resources, funding acquisition, writing – review & editing.

Conflicts of interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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References

- 1 S. Chae, W.-J. Kwak, K. S. Han, S. Li, M. H. Engelhard, J. Hu, C. Wang, X. Li and J.-G. Zhang, *ACS Energy Lett.*, 2021, **6**, 387–394.
- 2 T. Kim, W. Song, D.-Y. Son, L. K. Ono and Y. Qi, *J. Mater. Chem. A*, 2019, **7**, 2942–2964.

- 3 Y. Liu, Y. Zhu and Y. Cui, *Nat. Energy*, 2019, **4**, 540–550.
- 4 N. Kim, S. Chae, J. Ma, M. Ko and J. Cho, *Nat. Commun.*, 2017, **8**, 812.
- 5 X. Chen, H. Li, Z. Yan, F. Cheng and J. Chen, *Sci. China Mater.*, 2019, **62**, 1515–1536.
- 6 J. K. Lee, C. Oh, N. Kim, J.-Y. Hwang and Y.-K. Sun, *J. Mater. Chem. A*, 2016, **4**, 5366–5384.
- 7 B. Zhu, G. Liu, G. Lv and Y. Mu, *Sci. Adv.*, 2019, **5**, 0651.
- 8 K. Yao, N. Li, N. Li, E. Sivonxay, Y. Du, K. A. Persson, D. Su and W. Tong, *Chem. Mater.*, 2023, **35**, 2281–2288.
- 9 Z. Yang, Z. Li, Y. Yang, Q. Zhang, H. Xie, J. Wang, K. Świerczek and H. Zhao, *ACS Nano*, 2023, **17**, 7806–7812.
- 10 S. Mei, B. Xiang, S. Guo, J. Deng, J. Fu, X. Zhang, Y. Zheng, B. Gao, K. Huo and P. K. Chu, *Adv. Funct. Mater.*, 2023, **34**, 2301217.
- 11 Z. Li, M. Han, P. Yu, Q. Wu, Y. Zhang and J. Yu, *J. Energy Storage*, 2023, **67**, 107582.
- 12 D. J. Chung, D. Youn, J. Y. Kim, W. J. Jeong, S. Kim, D. Ma, T. R. Lee, S. T. Kim and H. Kim, *Small*, 2022, **18**, 2202209.
- 13 Z. Y. Zhang, Z. W. Li, Q. Luo, B. Z. Yang, Y. Liu, Y. Y. Hu, X. B. Liu, Y. H. Yin, Y. S. Li and Z. P. Wu, *Carbon*, 2022, **188**, 238–245.
- 14 B. H. Park, J. H. Jeong, G.-W. Lee, Y.-H. Kim, K. C. Roh and K.-B. Kim, *J. Power Sources*, 2018, **394**, 94–101.
- 15 M. Cai, Z. Zhao, X. Qu, J. Qu, Z. Hu, H. Shi, S. Gao, D. Wang and H. Yin, *Energy Storage Mater.*, 2023, **57**, 568–576.
- 16 H. Su, X. Li, C. Liu, Y. Shang and H. Liu, *Chem. Eng. J.*, 2023, **451**, 138394.
- 17 Q. Shi, J. Zhou, S. Ullah, X. Yang, K. Tokarska, B. Trzebicka, H. Q. Ta and M. H. Rummeli, *Energy Storage Mater.*, 2021, **34**, 735–754.
- 18 Y. Lu, Z. Ye, Y. Zhao, Q. Li, M. He, C. Bai, X. Wang, Y. Han, X. Wan, S. Zhang, Y. Ma and Y. Chen, *Carbon*, 2023, **201**, 962–971.
- 19 H. Luo, X. Zhang, Z. Wang, L. Zhang, C. Xu, S. Huang, W. Pan, W. Cai and Y. Zhang, *ACS Appl. Mater. Interfaces*, 2023, **15**, 4166–4174.
- 20 W. Li, Q. Ma, P. Shen, Y. Zhou, L. Soule, Y. Li, Y. Wu, H. Zhang and M. Liu, *Nano Energy*, 2021, **80**, 105506.
- 21 G. Coquil, J. Fullenwarth, G. Grinbom, M. T. Sougrati, L. Stievano, D. Zitoun and L. Monconduit, *J. Power Sources*, 2017, **372**, 196–203.
- 22 Z. Teng, X. Su, Y. Zheng, J. Sun, G. Chen, C. Tian, J. Wang, H. Li, Y. Zhao and G. Lu, *Chem. Mater.*, 2012, **25**, 98–105.
- 23 K. Li, G. Yuan, X. Liu, Y. Guo, R. Huang, H. Li, H. Zhang, Q. Jia, Z. Xie, S. Zhang and W. Lei, *Adv. Funct. Mater.*, 2023, **33**, 2302348.
- 24 Z. Lu, L. Zhang and X. Liu, *J. Power Sources*, 2010, **195**, 4304–4307.
- 25 Y. He, X. Yu, Y. Wang, H. Li and X. Huang, *Adv. Mater.*, 2011, **23**, 4938–4941.
- 26 B. Li, S. Yang, S. Li, B. Wang and J. Liu, *Adv. Energy Mater.*, 2015, **5**, 1500289.
- 27 F. M. Hassan, R. Batmaz, J. Li, X. Wang, X. Xiao, A. Yu and Z. Chen, *Nat. Commun.*, 2015, **6**, 8597.
- 28 Z. Sun, J. Zhu, C. Yang, Q. Xie, Y. Jiang, K. Wang and M. Jiang, *ACS Appl. Mater. Interfaces*, 2023, **15**, 12946–12956.
- 29 Y. Sun, J. Lopez, H. W. Lee, N. Liu, G. Zheng, C. L. Wu, J. Sun, W. Liu, J. W. Chung, Z. Bao and Y. Cui, *Adv. Mater.*, 2016, **28**, 2455–2461.
- 30 J. Long, W. He, H. Liao, W. Ye, H. Dou and X. Zhang, *ACS Appl. Mater. Interfaces*, 2023, **15**, 10726–10734.
- 31 X. Jiao, J. Yin, X. Xu, J. Wang, Y. Liu, S. Xiong, Q. Zhang and J. Song, *Adv. Funct. Mater.*, 2020, **31**, 2005699.
- 32 P. Xiao, X. Yun, Y. Chen, X. Guo, P. Gao, G. Zhou and C. Zheng, *Chem. Soc. Rev.*, 2023, **52**, 5255–5316.
- 33 J. Lin, L. Wang, Q. Xie, Q. Luo, D. L. Peng, C. Buddie Mullins and A. Heller, *Angew. Chem., Int. Ed.*, 2023, **62**, e202216557.
- 34 Z. Piao, R. Gao, Y. Liu, G. Zhou and H. M. Cheng, *Adv. Mater.*, 2023, **35**, 2206009.
- 35 Z. Piao, X. Wu, H.-R. Ren, G. Lu, R. Gao, G. Zhou and H.-M. Cheng, *J. Am. Chem. Soc.*, 2023, **145**, 24260–24271.
- 36 X. Wu, Z. Piao, M. Zhang, G. Lu, C. Li, K. Jia, Z. Zhuang, R. Gao and G. Zhou, *J. Am. Chem. Soc.*, 2024, **146**, 14036–14047.
- 37 G. Zeng, Y. An, S. Xiong and J. Feng, *ACS Appl. Mater. Interfaces*, 2019, **11**, 23229–23235.
- 38 X. Liu, R. Wang, S. Liu, J. Pu, H. Xie, M. Wu, D. Liu, Y. Li and J. Liu, *Adv. Energy Mater.*, 2023, **13**, 2302987.
- 39 J. Zheng, J. Liu, S. Wang, F. Luo, L. Ben and H. Li, *Chin. Phys. B*, 2020, **29**, 048203.
- 40 C. Xu, F. Lindgren, B. Philippe, M. Gorgoi, F. Björefors, K. Edström and T. Gustafsson, *Chem. Mater.*, 2015, **27**, 2591–2599.
- 41 L. Gehrlein, C. Njel, F. Jeschull and J. Maibach, *ACS Appl. Energy Mater.*, 2022, **5**, 10710–10720.
- 42 N.-S. Choi, K. H. Yew, K. Y. Lee, M. Sung, H. Kim and S.-S. Kim, *J. Power Sources*, 2006, **161**, 1254–1259.
- 43 S. Yamazaki, R. Tatara, H. Mizuta, K. Kawano, S. Yasuno and S. Komaba, *J. Phys. Chem. C*, 2023, **127**, 14030–14040.
- 44 A. Mukanova, A. Serikkazyeva, A. Nurpeissova, S.-S. Kim, M. Myronov and Z. Bakenov, *Electrochim. Acta*, 2020, **330**, 800–806.
- 45 Y. Jin, N.-J. H. Kneusels, L. E. Marbella, E. Castillo-Martínez, P. C. M. Magusin, R. S. Weatherup, E. Jönsson, T. Liu, S. Paul and C. P. Grey, *J. Am. Chem. Soc.*, 2018, **140**, 9854–9867.
- 46 Y. Yang, Z. Yang, Y. Xu, Z. Li, N. Yao, J. Wang, Z. Feng, K. Wang, J. Xie and H. Zhao, *J. Power Sources*, 2021, **514**, 2302068.
- 47 Y. Kamikawa, K. Amezawa and K. Terada, *J. Phys. Chem. C*, 2020, **124**, 19937–19944.
- 48 M. Salah, T. Pathirana, E. A. de Eulate, C. Hall, R. Kerr and M. Fabretto, *J. Energy Storage*, 2022, **46**, 12243–12256.
- 49 L. Chen, K. Wang, X. Xie and J. Xie, *J. Power Sources*, 2007, **174**, 538–543.
- 50 R. Nölle, A. J. Achazi, P. Kaghazchi, M. Winter and T. Placke, *ACS Appl. Mater. Interfaces*, 2018, **10**, 28187–28198.
- 51 C. Sun, X. Zhang, C. Li, K. Wang, X. Sun, F. Liu, Z.-S. Wu and Y. Ma, *J. Energy Chem.*, 2022, **64**, 442–450.
- 52 Z. Cao, X. Zheng, Q. Qu, Y. Huang and H. Zheng, *Adv. Mater.*, 2021, **33**, 2103178.
- 53 F. Aupperle, G. G. Eshetu, K. W. Eberman, A. Xioa, J.-S. Bridel and E. Figgemeier, *J. Mater. Chem. A*, 2020, **8**, 19573–19587.

- 54 G. Yang, S. Frisco, R. Tao, N. Philip, T. H. Bennett, C. Stetson, J.-G. Zhang, S.-D. Han, G. Teeter, S. P. Harvey, Y. Zhang, G. M. Veith and J. Nanda, *ACS Energy Lett.*, 2021, **6**, 1684–1693.
- 55 Q. Li, X. Liu, X. Han, Y. Xiang, G. Zhong, J. Wang, B. Zheng, J. Zhou and Y. Yang, *ACS Appl. Mater. Interfaces*, 2019, **11**, 14066–14075.
- 56 J. Cheng, Z. Huang, A. Lu, A. He, Y. Shao, Y. Fan and Y. Huang, *Journal of Materiomics*, 2025, **11**, 100941.
- 57 F. Weigend and R. Ahlrichs, *Phys. Chem. Chem. Phys.*, 2005, **7**, 3297–3305.
- 58 T. Lu and F. Chen, *J. Comput. Chem.*, 2011, **33**, 580–592.
- 59 L. K. Zhao, X. W. Gao, J. Mu, W. B. Luo, Z. Liu, Z. Sun, Q. F. Gu and F. Li, *Adv. Funct. Mater.*, 2023, **33**, 2304292.
- 60 L.-K. Zhao, X.-W. Gao, Q. Gu, X. Ge, Z. Ding, Z. Liu and W.-B. Luo, *eScience*, 2024, **4**, 100201.
- 61 S. Chen, G. Zheng, X. Yao, J. Xiao, W. Zhao, K. Li, J. Fang, Z. Jiang, Y. Huang, Y. Ji, K. Yang, Z.-W. Yin, M. Zhang, F. Pan and L. Yang, *ACS Nano*, 2024, **18**, 6600–6611.
- 62 D.-H. Kim, B. Kang and H. Lee, *J. Power Sources*, 2019, **423**, 137–143.
- 63 J. K. Lee, W. Y. Yoon and B. K. Kim, *J. Electrochem. Soc.*, 2014, **161**, A927–A933.
- 64 N.-Y. Park, G.-T. Park, S.-B. Kim, W. Jung, B.-C. Park and Y.-K. Sun, *ACS Energy Lett.*, 2022, **7**, 2362–2369.
- 65 A. L. Michan, B. S. Parimalam, M. Leskes, R. N. Kerber, T. Yoon, C. P. Grey and B. L. Lucht, *Chem. Mater.*, 2016, **28**, 8149–8159.
- 66 S. Zhang, *Electrochem. Commun.*, 2003, **5**, 979–982.
- 67 G. Wen, L. Tan, X. Lan, H. Zhang, R. Hu, B. Yuan, J. Liu and M. Zhu, *Sci. China Mater.*, 2021, **64**, 2683–2696.
- 68 Z. Deng, Z. Huang, Y. Shen, Y. Huang, H. Ding, A. Luscombe, M. Johnson, J. E. Harlow, R. Gauthier and J. R. Dahn, *Joule*, 2020, **4**, 2017–2029.
- 69 S. J. An, J. Li, C. Daniel, D. Mohanty, S. Nagpure and D. L. Wood, *Carbon*, 2016, **105**, 52–76.